

Offshore Workover Operations: Reducing Uncertainty of Critical Weather Scenarios by Optimal Use of Simulations and Probabilistic Machine Learning

Simen Eldevik

Group Technology and Research, DNV GL, Oslo, Norway. E-mail: simen.eldevik@dnvgl.com

Stian Sætre

Riser Technology, DNV GL, Oslo, Norway.

Offshore workover operations are expensive maritime operations that are highly affected by the stochastic nature of sea-state and weather in combination with the structural integrity of the asset. This often cause delays and postponement or aborted operations which incur additional costs of several hundred thousand USD per day. The operator needs to assess the current weather and ocean forecast and its potential to interrupt the planned drilling operation continuously. For many weather scenarios the integrity of the riser system and wellhead is the limiting factor, and accurate estimation of the capacity and utilization of these are critical to optimal and safe operation.

A common approach is to run a set of simulations prior to operations that establish the relation between the environmental loads, the response of the rig and the response and capacity of the riser-well system. This approach is often based on a factorial approach of scenario selection and tend to spend a lot of simulation effort on non-consequential scenarios. This paper focus on how Gaussian Process (GP) regression, a probabilistic machine learning method, can be a better alternative to guide simulation efforts towards scenarios critical for the decision context. It also shows how a GP model can be used to improve uncertainty in live decision support during operations.

Keywords: Offshore drilling, Weather window, Simulations, Design-of-experiments, Probabilistic machine learning, Gaussian process regression, Uncertainty.

1. Introduction

The day rate of offshore drill rigs are several hundred thousand USD per day. Thus, optimized use of potential rig time is essential to reduce operational cost. The limiting factor of whether a drilling operation can continue or if it needs to be postponed or aborted is often governed by the weather and sea-state the drill rig operates in. The operators have limited time between a weather and ocean forecast is available and when a decision on continued or aborted operation is needed. To avoid unnecessary interruptions, it is essential to be able to acquire accurate information about the utilization of the workover equipment within the time-frame the weather forecast permits. For benign or severe forecasts, the decision is often trivial and can be made based on past experiences and simple operational limits. However, for many weather scenarios the structural integrity of the riser system and wellhead is the limiting factor, and accurate estimation of the capacity and utilization of these are critical to optimal and safe operation.

This assessment is a complex process where specific information about the rig, the riser system, the wellhead, and environmental conditions need to be combined through several dynamic

models. One model establishes the relation between weather, sea-states and rig movement, while another model analyses the structural integrity and utilization of the riser and wellhead given the dynamic response of the rig. Because of the stochastic nature of how the weather, e.g. wave loads, current and wind, affect the rig's dynamic position and consequently the riser system's integrity, it is computationally challenging to provide decision support based on real-time simulations of all relevant load scenarios. A common approach is to run a set of simulations prior to operations that establish the relation between the environmental loads, the response of the rig and the response and capacity of the riser-well system. Then, during operation these results are often used as a look-up to compare current weather forecast with the simulated utilization.

In this paper we look at the operator's decision to continue or abort a drilling operation based on weather forecasts. The paper illustrates how probabilistic machine learning, can be used to optimize the simulation effort to minimize uncertainty in weather scenarios that are critical for the operator's decision context.

2. Problem description

To assess the structural integrity of the riser stack and wellhead for relevant weather scenarios, the operator needs access to advanced and time-consuming simulation models. For all relevant sea-states (e.g. wave loads and current) the dynamic movement of the drill rig and its ability to maintain position (e.g. through dynamic positioning or mooring) needs to be simulated. Based on the resulting dynamic motion, another advanced and time-consuming simulation must be run to assess the structural response of the riser system and the forces it exert on the wellhead. The utilization u of the riser wellhead connection's capacity is treated as an unknown function that this simulation chain seeks to estimate.

$$u = f(\mathbf{x}) \quad (1)$$

The relevant parameters $\mathbf{x}$ that dominate this utilization is: The sea-state, described by significant wave height, H_s , wave period, T_p , the current velocity, v_c , the dynamic position and off-set from the centered vessel position, d , and the internal pressure of the riser, p_{int} .

The function f is a mapping of the five-dimensional input space to the one-dimensional output space, $f : \mathbb{R}^5 \rightarrow \mathbb{R}$, and the challenge is to explore this multi-dimensional input space sufficiently to have the necessary confidence in the estimated utilization for all relevant operation- and weather-scenarios.

3. Surrogate Models and Emulators

If the function f is expensive or time-consuming, we cannot practically evaluate $f(\mathbf{x})$ at every point $\mathbf{x}$ within the time-frame of the relevant decision context. Because we cannot evaluate it as often or in real-time as needed, we need to construct a fast-running approximation to the real response away from where it has been evaluated. Such a fast approximation of a real process, $\hat{f} \approx f$, based on a finite set of observed (training) data, $\{f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)\}$, is often referred to as a surrogate model, and is what most machine learning (ML) algorithms produce. However, to use such a surrogate model to support safety-critical decisions in a complex system, good and efficient uncertainty analysis and sensitivity analysis is vital. O'Hagan (2006) argues that to achieve this, the surrogate model should also be able to interpolate, i.e. ensure that all approximations match the observations exactly at the training points, and it should be able to provide an uncertainty estimate of the prediction on all points that are not part of the training set. These two traits are typically not available for many ML algorithms), and O'Hagan refer to surrogate models having these traits as

emulators.

Emulator is a surrogate model that can:

- Interpolate, i.e. $\hat{f}(\mathbf{x}_i) = f(\mathbf{x}_i)$ for all training data $\{\mathbf{x}_i\}_{i=1}^N$
- Provide uncertainty estimates on $\hat{f}(\mathbf{x})$ for all data not in the training set $\mathbf{x} \notin \{\mathbf{x}_i\}_{i=1}^N$

To have an emulator that can be used for decision support we need to explore the state-space prior to the decision based on a finite set of evaluations or observations of the complex system.

3.1. Adaptive exploration of the state-space

Without prior knowledge of how these parameters act together to affect the utilization of the riser and wellhead system, the engineers often decide on a factorial approach Fisher (1937) that runs through all variations of each level of the variables to establish the response space. Although this is known to be an inferior approach to explore a state space it is nevertheless a prevalent method among modelers because of its simplicity Saltelli and Annoni (2010). This results in an exploration that needs to cover the entire state space and is ignorant of the critical limit of the response. Thus, a significant amount of the simulation efforts may prove, after the fact, to have been spent in areas of the state space that have an insignificant effect on the decision problem. A typical example of this is where a lot of the simulation effort has been spent where the system is well within the critical operational limit, or far outside where the system would have been disconnected because it is clear that it would not have survived those conditions. A more efficient way to explore a state space is by space-filling sampling strategies like e.g. Latin Hypercube Sampling, LHS McKay et al. (1979) or a Sobol sequence Sobol (1976). However, these strategies are also prone to spending more of the simulation effort in areas which is not critical for the decision context. Thus, the optimal way would be to start off with a smaller space filling sample, and then adaptively select new simulation scenarios based on the acquired knowledge of the response and some objective MacKay (1992).

3.2. Probabilistic ML - Gaussian process

As we are searching for an unknown function that describes the response of the system it makes sense to consider inference directly in function space. To do this we use a Gaussian Process (GP) to describe a distribution over functions. A Gaussian Process is completely specified by its mean function and covariance function. Rasmussen and Williams Rasmussen and Williams

(2006) define the mean and covariance function of a real process $f(\mathbf{x})$ as

$$m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})] \quad (2a)$$

$$k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))] \quad (2b)$$

and write the Gaussian Process as

$$f \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')). \quad (3)$$

For any finite collection $[\mathbf{x}_1, \dots, \mathbf{x}_n]$ of points $\mathbf{x}_i \in \mathbb{R}^D$, the distribution of $[f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)]$ is multivariate Gaussian with mean $m(\mathbf{x}_i)$ and covariance $\mathbf{K}_{i,j} = k(\mathbf{x}_i, \mathbf{x}_j)$ for $i, j \in [1, n]$. If a set of random variables are jointly multivariate normally distributed, then the marginal distribution of subsets of the random variables will also be Gaussian.

Thus, if we have a data set $\mathcal{D}$ of N noisy observations of the real process, i.e. $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$ where $y_i = f(\mathbf{x}_i) + \epsilon_i$, and we assume the noise term ϵ_i is independent and identically normally distributed with zero-mean and common variance c^2 , this can be used to infer the distribution of $f(\mathbf{x}^*)|\mathcal{D}$ at unobserved points $\mathbf{x}^*$ in the input space.

The predictive posterior distribution at n new inputs

$\mathbf{f}^*|\mathcal{D} = [f(\mathbf{x}_1^*), \dots, f(\mathbf{x}_n^*)] | \{y_i = f(\mathbf{x}_i) + \epsilon_i\}_{i=1}^N$ is then given in closed form as

$$\mathbf{f}^*|\mathcal{D} \sim N(\mu_{\mathbf{f}^*|\mathcal{D}}, \Sigma_{\mathbf{f}^*|\mathcal{D}}) \quad (4)$$

with mean and covariance

$$\mu_{\mathbf{f}^*|\mathcal{D}} = \mathbf{m}_* + \mathbf{K}_* (\mathbf{K}_* + c^2 \mathbf{I})^{-1} (\mathbf{y} - \mathbf{m}) \quad (5a)$$

$$\Sigma_{\mathbf{f}^*|\mathcal{D}} = \mathbf{K}_{**} - \mathbf{K}_* (\mathbf{K}_* + c^2 \mathbf{I})^{-1} \mathbf{K}_*^T \quad (5b)$$

where $\mathbf{m}_*$, $\mathbf{m}$ and $\mathbf{y}$ are vectors with elements $m(\mathbf{x}_i^*)$, $m(\mathbf{x}_i)$ and y_i respectively. $\mathbf{I}$ is the $N \times N$ identity matrix, and $\mathbf{K}_*$ and $\mathbf{K}_{**}$ have elements $(\mathbf{K}_*)_{i,j} = k(\mathbf{x}_i^*, \mathbf{x}_j)$ and $(\mathbf{K}_{**})_{i,j} = k(\mathbf{x}_i^*, \mathbf{x}_j^*)$ respectively.

For the current application we want the GP to predict the utilization of critical components in the riser stack up during different weather conditions. In this example the critical component has a sufficient structural integrity if the combination of effective tension and bending moment are below the component's tension- and bending capacity. A utilization function of this capacity can then be defined as the critical decision parameter, and here a utilization of 1.0 is the critical limit for when a drilling operation needs to be aborted or postponed, and thus, we select a prior mean function $m(\mathbf{x}) = 1.0$. The covariance function k governs how much the function can vary between two points that are close to each other. Here we select

the Matérn 3/2 kernel, which guarantees that the function is once differentiable, and is defined by

$$k(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \sqrt{3}r\right) \exp\left(-\sqrt{3}r\right), \quad (6)$$

where

$$r = \sqrt{\sum_{i=1}^D \frac{(\mathbf{x}_i - \mathbf{x}'_i)^2}{l_i^2}}. \quad (7)$$

σ is the kernel variance and l_i is the length scale in dimension i . The GP model is trained by estimating its hyperparameters, the kernel variance σ , the length scale parameters l_i and the white noise variance c , based on the training data, typically by maximum likelihood or Bayesian methods Rasmussen and Williams (2006).

In this work, the Pyro: Deep Universal Probabilistic Programming language Bingham et al. (2018) has been used to train the GP regression model and make predictions according to Eq. (5a) and Eq. (5b).

4. Optimize according to decision context

To be able to optimally select simulation scenarios we need to define an objective to optimize. As the decision to postpone or abort a drilling operation is dominated by the structural utilization of critical components within the riser stack up, we want to minimize the uncertainty in our prediction close to this critical decision boundary. This means that we want to focus the simulation effort close to where the system response results in a utilization of 1.0. Without any prior information about the underlying response function an initial space filling design will give a good starting point for further optimal exploration-exploitation strategies. The Bayesian nature of a GP model makes it suitable for exploration-exploitation strategies based on uncertainty reduction. For each possible point in the input space the GP model has an associated mean and variance. One intuitive way to explore the input space while exploiting the information already gained, is to select a new simulation scenario at the input combination where the GP model has the largest variance (i.e. local uncertainty reduction MacKay (1992)). It is also possible to reduce the overall uncertainty of the entire input space Cohn (1996). However, these methods are most suitable when the objective is to get a good approximation of the real response for the entire possible input space.

For the problem at hand, it is important to find the boundary between acceptable and unacceptable operating conditions. We do not want the GP model to support a decision to abort an operation when in fact it could have continued safely, but more importantly, we cannot let the

GP model support a decision to continue operation when in fact it will lead to a structural failure of the riser-well system. Thus, the objective is to reduce the uncertainty close to this decision boundary between continued and aborted drilling operations, which is decided by the utilization function crossing the threshold $u = 1.0$.

To do this it is reasonable to select an exploration-exploitation strategy that focus on establishing the multidimensional contour line in the utilization response space as described by Ranjan et al. Ranjan et al. (2008). Such a method will focus simulation efforts in the vicinity of the critical response utilization $u = 1.0$. This also means that for very calm weather scenarios and sea-states the riser-well system is well within its structural capacity ($u \ll 1.0$) – and we do not need to spend a lot of simulation efforts for these scenarios. On the other hand, for very bad weather scenarios, the riser-well system will experience loads far above what it can withstand ($u \gg 1.0$), and thus we do not need to spend a lot of simulation effort for these scenarios either.

4.1. Comparing sampling strategies

To see the effect of an adaptive sampling strategy with an objective to reduce the uncertainty close to the critical decision boundary, we start by looking at common a priori sampling strategies. If we assume that the simulated response sufficiently represent the true response, we let the operator make decisions based on the predicted utilization from a GP model trained on simulations. In the following we denote a *Positive* prediction to be where the operator decide to commence or continue the drilling operation, while a *Negative* prediction is where the operator decide to postpone or abort the drilling operation. Under these assumptions the operator may experience four possible outcomes:

- (i) **True Positive**
If the prediction is below the critical utilization threshold (possibly by a conservative margin), the operator will continue operation, and if post facto the utilization was indeed below the threshold, this was the correct decision.
- (ii) **True Negative**
Similarly, if the prediction is above the critical utilization threshold the operator will postpone or abort the operation, and if ex post the simulation shows that this in fact was the case, this was also the correct decision.
- (iii) **False Positive**
The most critical erroneous decision, is where the prediction is below the threshold while the post facto utilization proves to be above the critical threshold. Such an erroneous decision compromise the integrity of the system, and is the outcome we have to guard against.

- (iv) **False Negative**

The other erroneous decision, where the prediction is above the threshold while the simulated utilization proves to be less than the threshold, is less critical as it does not compromise the integrity of the system. However, we want to minimize the rate of such erroneous decisions as it leads to unnecessary postponement or aborted drilling operations and added costs.

To compare the different sampling strategies and their efficacy we compare the rate of False Positive and False Negative predictions. An optimal solution is where we eliminate the False Positive predictions while at the same time minimize the False Negative predictions.

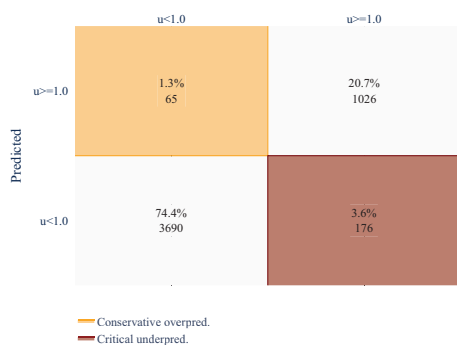
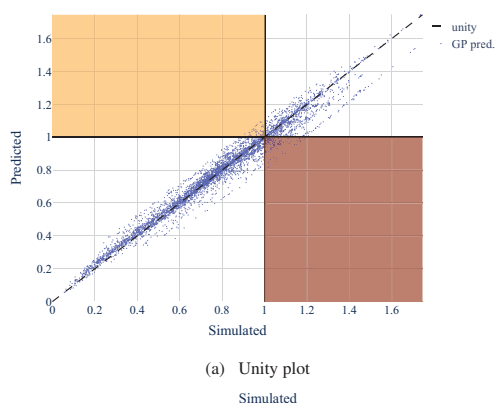
To train the GP model we allow 500 simulations for each sampling strategy while we test the efficacy of the trained GP model on 5000 randomly selected simulation scenarios.

4.1.1. Full factorial sampling

A full factorial sampling strategy is where we consider each factor of our input space (i.e. $x \in \mathbb{R}^5$) and divide each factors in a discrete number of possible values. The full factorial sampling strategy is then to loop through all possible combinations of these different levels across all factors Fisher (1937). Although this is known to be an inferior approach to explore a state space it is nevertheless a prevalent method among modelers because of its simplicity Saltelli and Annoni (2010). If we select 4 levels for the 5-dimensional input-space, this yields $N = 4^5 = 1024$ possible combinations. To adhere to the 500 allowed simulation scenarios we down-sample randomly from the 1024 full factorial combinations to 500 simulation scenarios that lie on the full factorial input grid.

Figure 1(a) shows the unity plot where the trained GP (here based on 500 full factorial simulations) have been used to predict the utilization of 5000 random simulation results. The GP prediction is plotted as blue dots on the y-axis versus the actual simulation result on the x-axis. If the GP prediction was perfect, all blue dots would lie on the dashed black unity line. Blue dots that fall below the unity line, signify actual simulated utilization that has been under-predicted by the GP model, while blue dots above the unity line signify actual simulated utilization that has been over-predicted by the GP model. Whether this under- or over-prediction is a critical False Positive or costly False Negative depends on the magnitude of the predicted utilization. It is also clear that far from the decision boundary $u = 1.0$, the decision supported by the GP is OK irrespective of a higher prediction uncertainty.

Figure 1(b) shows a confusion matrix where the decision boundary $u \geq 1.0$ have been applied.



(b) Confusion matrix

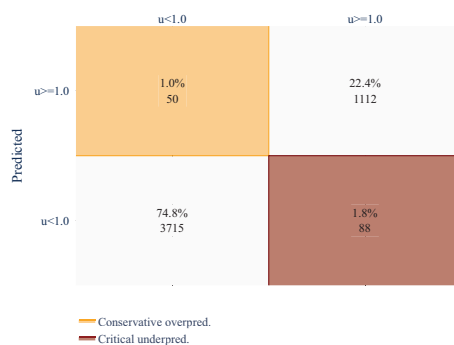
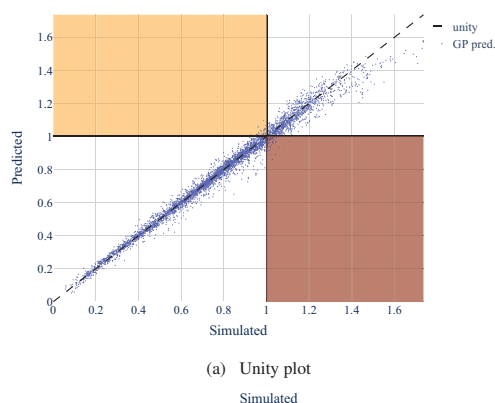
Fig. 1. Utilization prediction of 5000 random scenarios vs. actual simulation results from a GP model trained on 500 simulation scenarios of Full factorial samples.

It is clear from Fig. 1 that a significant amount of the 5000 test simulations are critically under-predicted. 3.6% (176) of the scenarios lead to a critically erroneous decisions (False Positives in dark red to the right), while 1.3% (65) of the scenarios lead to a costly under-prediction (False Negatives in orange on the top row).

4.1.2. Latin hypercube sampling

A more efficient way to explore a state space is by space-filling sampling strategies like e.g. Latin Hypercube Sampling, LHS McKay et al. (1979) or a Sobol sequence Sobol (1976). In Fig. 2 the unity plot (a) and confusion matrix (b) of the 5000 random scenarios when the GP model has been trained on 500 simulations from a LHS strategy. The LHS strategy reduce the critical False Positive rate to 1.8% (88), while the costly False Negatives are reduced to 1.0% (50) of the test scenarios.

Because of the Bayesian nature of the GP model it is possible to reduce the critical False Positive



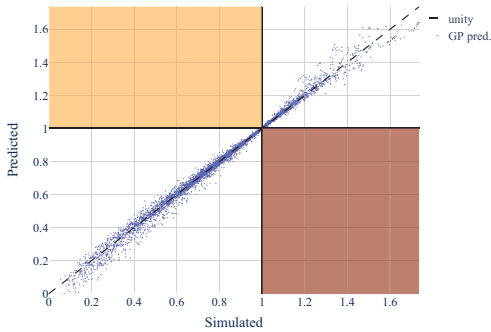
(b) Confusion matrix

Fig. 2. Utilization prediction of 5000 random scenarios vs. actual simulation results from a GP model trained on 500 simulation scenarios of Latin Hypercube samples.

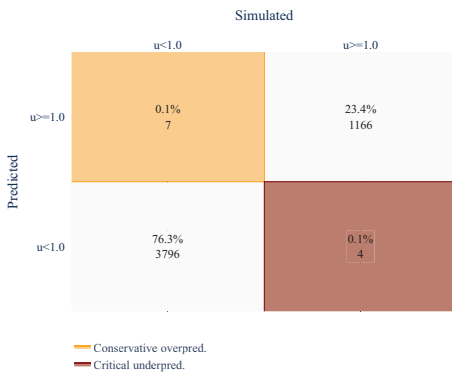
rate for the full factorial and LHS strategies by using an upper-bound confidence level of the GP prediction. Adding a factor of the variance in the GP prediction would move all the predictions up in the unity plot and confusion matrix. However, then these strategies would be punished significantly by much higher False Negative rates.

4.1.3. Adaptive sampling

The adaptive sampling strategy utilized here focus the simulation effort close to the critical decision boundary (i.e. $u \sim 1.0$). The first 100 scenarios was selected by a LHS approach. Then, through 80 cycles of adaptive learning, 5 candidate simulations were selected in each cycle based on an exploration-exploitation strategy to maximize the information gain around the critical utilization contour Ranjan et al. (2008). As can be seen in Fig. 3(a), the spread in blue dots around the unity line significantly narrows close to the critical decision boundary. This shows that the adaptive ap-



(a) Unity plot



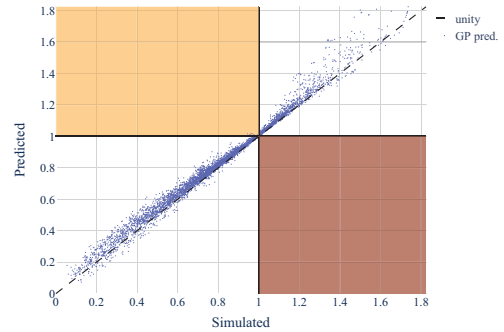
(b) Confusion matrix

Fig. 3. Utilization prediction of 5000 random scenarios vs. actual simulation results from a GP model trained on 100 initial LHS scenarios and 400 AL scenarios (80 cycles of 5).

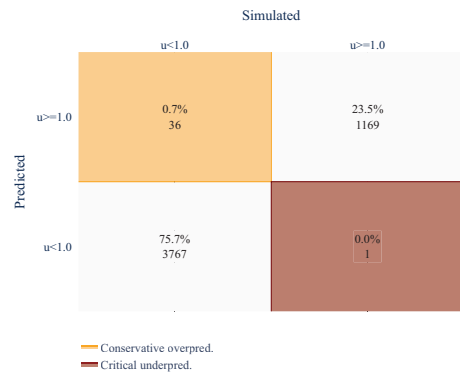
proach significantly reduce the uncertainty related to erroneous decisions based on predictions by this GP model. The spread in predictions for low or high utilizations are higher, but because of the decision boundary this uncertainty is insignificant with respect to the relevant decision context.

Figure 3(b) shows that the critical under-prediction rate has fallen to 0.1% (4) and the over-prediction rate is 0.1% (7). This is a significant improvement compared to the a priori strategies presented above.

Because the adaptive approach used here reduce the prediction uncertainty close to the decision boundary, but not necessarily far from this boundary, it is possible to even further reduce the critical under-predictions without being punished significantly by much higher False Negative rates. In Fig. 4 the upper 95% confidence bound has been used instead of the mean prediction from the GP model. As can be seen by comparing Fig. 3(a) and Fig. 4(a), all the blue dots have been moved



(a) Unity plot



(b) Confusion matrix

Fig. 4. Utilization prediction of 5000 random scenarios vs. actual simulation results from the AL trained GP model adjusted to upper 95% confidence level.

up in the latter. However, because the uncertainty in the GP prediction is much less close to where the simulation effort has been focused (i.e. around $u = 1.0$) this shift is larger away from the decision boundary – where it does not affect the decision. Thus, one can reduce the False Positive rate close to $\sim 0\%$ (1), while not being punished by a significant increase in False Negatives which still is only 0.7% (36).

5. Conclusive remarks

This work has illustrated and shown how probabilistic machine learning models, like Gaussian process regression, and an adaptive approach to selection of simulation scenarios can drastically improve the efficiency and efficacy when applied to inform safety-critical decisions. It has also shown that the Bayesian nature of probabilistic models have an advantageous effect when uncertainty in the prediction is critical for the decision context. By minimizing uncertainty where

uncertainty have the highest impact on the outcome, it is possible to confidently reduce the rate of unwanted critical outcomes without the usual negative effects of being too conservative.

To ensure safe operations of safety-critical systems where decisions are supported by machine-learned models, it is imperative that incorporate a measure of risk. They need to be able to quantify the risk of making erroneous predictions Eldevik (2018), and thus the Bayesian nature of GP models and their ability to quantify uncertainty is a key advantage.

Note however, that, even though the 5000 scenarios used to test the model shows a reasonable uncertainty level for all the sampled weather scenarios above, this is no guarantee that the model is sufficiently good for all possible future weather states. The Bayesian nature of GP models make it easy to quickly update the prediction model given more observation/evidence. The operator of drilling rigs are subjected to a highly dynamic decision context and need to continuously assess how the forecasted weather scenarios may affect the integrity of the riser wellhead system. By combining a GP model as illustrated in this work with online forecast services and on-demand simulation capabilities, it can act as a real-time tool for decision-support with the added capability to suggest which new simulation scenarios should be run to reduce prediction uncertainty on-demand. An example of how this can be implemented is documented in Eldevik et al. (2020).

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